

An 157TOPS/W Transformer Learning Processor Supporting Forward Pass Only with Zeroth-Order Optimization

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Abstract

This work presents the *first* Transformer learning processor that supports forward pass only with zeroth-order optimization. By applying uniform perturbation, sign-based weight gradient generation, and exponential ternary speculation, the training complexity is reduced by up to 96%. By employing parallel processing, tiled-based bitmask generation, and workload scheduling, the latency is reduced by up to 96%. Fabricated in 40-nm CMOS, the chip achieves a max energy efficiency of 157TOPS/W, outperforming prior backpropagation-based Transformer learning processors by 1.6-to-7.6 $\times$.

Introduction

On-device learning has become essential for privacy preserving for AI model adaptation. Conventionally, backpropagation-based learning [2], including forward pass and backward pass, is employed. As Transformer-based neural network [1] has become prominence, hardware acceleration has gained great attention and Transformer learning processors have been demonstrated in [3, 4]. Fig. 1 shows the overview of the learning algorithms. For the BP-based algorithm, sequential execution on forward pass and backward pass limits acceleration. As the sequence length increases, the learning complexity increases dramatically. As an example, backward pass exhibits 12 $\times$ higher computational complexity than forward pass for a sequence length of 2048. The forward pass only learning algorithm [5] has been introduced without the need for backward pass, but hardware implementation has not been investigated. This work presents the first Transformer learning processor supporting forward pass only with zeroth-order optimization.

Fig. 2 shows the workflow of the forward pass only learning algorithm. Unlike the backpropagation-based algorithm, which computes gradients by propagating error backward, the forward pass only-based algorithm obtains the gradient from the loss function directly [5]. This reduces the computational complexity by 85% for a sequence length of 2048. In this work, the computational complexity can be further reduced by applying three techniques: uniform perturbation, sign-based weight gradient generation, and exponential ternary speculation. First, the zeroth-order gradients are generated by evaluating the loss function on randomly perturbed direction vectors. The Gaussian direction vector can be replaced with a random number with uniform distribution. Second, zeroth-order optimization originally estimates the gradient with the difference in evaluating the loss function. The weight gradients are generated in a sign-based manner. Third, exponential ternary speculation (ETS) is proposed to remove the redundant KV tokens by leveraging the characteristic of the SoftMax function. The approximated Key is generated by evaluating the exponent part of the input token and weight. The approximated Query is approximated by ternary representation. A speculated bitmask is then created to skip computations for weakly-related tokens. Overall, the computational complexity is reduced by 96%.

System Architecture

Fig. 3 shows the system architecture of the proposed Transformer learning processor, including a sparse data processing core (SDPC), a compute engine, and a tensor engine. In this work, the compute engine and the tensor engine are operated in parallel through workload scheduling. The SDPC is designed for generating bitmap and encoding the data. The compute engine performs matrix multiplications based on the encoded data. The tensor engine is designed to perform non-linear function and to generate weight gradients.

Fig. 4 shows the speculation unit in the SDPC that supports the proposed ETS. The exponent-only format allows to perform multiply-accumulate (MAC) operations with shift-and-add operations. The ternary format allows to simplify the PE array with multiplexers for multiplication. Conventionally, the bitmap encoding is implemented by uni-directional network [3], which consists of a large array of multiplexers and adders. In this work, a sparse encoding unit is designed for bitmap encoding with simple AND/OR gates and comparators, achieving a 57% power reduction. A bitmask is generated through tiling in IETS and the tiled bitmask is then applied to efficient attention computation, reducing the overall latency by 73%.

Fig. 5 shows the detailed architecture of the compute engine. Perturbation with random direction vectors is performed when weights are loaded to compute engine concurrently. Two processing cores are designed for computing two network outputs in parallel, reducing the latency by 70%. The compute engine can be operated in either the broadcasting mode or the unicast mode. Broadcasting is adopted for low data sparsity and the fetched data can be distributed without buffering, reducing the memory access by 75%. Unicasting is utilized for unbalanced loads and the dense non-zero data are sent to the corresponding MAC unit directly by index matching, increasing the utilization by up to 88%. The block floating-point (FP) arithmetic (with 1b sign, 7b mantissa, and 5b exponent) can be applied to the MAC unit [3], but there may exist redundant partial products when applying block FP arithmetic (due to shift operations on mantissa for data alignment). In this work, an 8-bit integer is split into upper and lower 4-bit parts. A 4-bit MAC unit with a leading one detector is proposed to skip zero-valued computations conditionally, reducing the energy by 20%. The computation flow for the attention layer is rescheduled, reducing the processing latency in the tensor engine by 49%. Overall, the latency is reduced by 96%.

Fig. 6 shows the detailed architecture of the special function unit in the tensor engine, which supports piecewise affine operations [6] for non-linear functions. The change of base formula with linear approximation is employed for the exponential and logarithmic operations. FP multiplication is approximated by adding the FP numbers as in the fixed-point format, in the logarithmic domain. The proposed design reduces the power and area of the special function unit by 66% and 79%, respectively. Hardware sharing is employed for row-wise and element-wise non-linear operations, reducing the area by 50%.

Experimental Results and Conclusion

Fig. 7 shows the chip implementation and measurement results. Fabricated in 40-nm CMOS, the proposed learning processor integrates 7.9M gates in core area 7.09 mm². The chip dissipates 16.2-to-229.3mW at 10-to-200MHz from a 0.58-to-1.02V supply. It achieves a maximum energy efficiency of 157TOPS/W at 50MHz, 0.58V. Fig. 8 shows the performance evaluation and comparison with the state-of-the-art designs [3, 4]. The chip achieves a $\leq 1.2\%$ performance loss on commonly-used BERT [7] and RoBERTa [8] models, when compared with the BP-based implementation, for natural language processing tasks. This work achieves 1.6-to-7.6 $\times$ higher energy efficiency than prior backpropagation-based designs.

Acknowledgements

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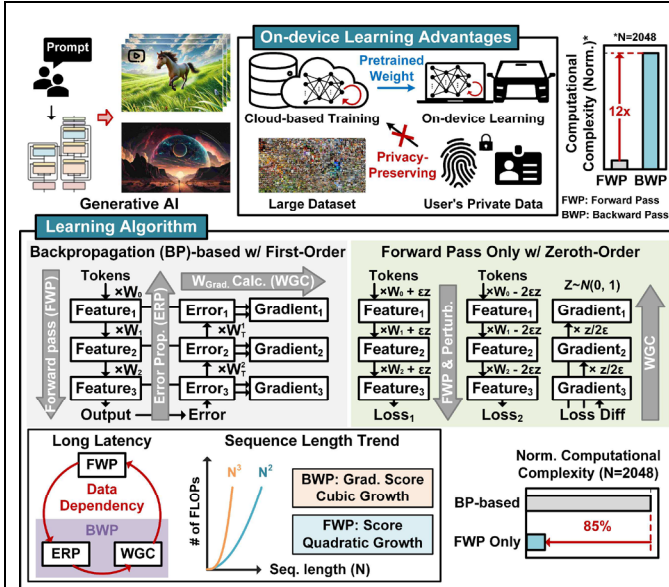


Fig. 1. On-device learning with Transformer-based neural networks.

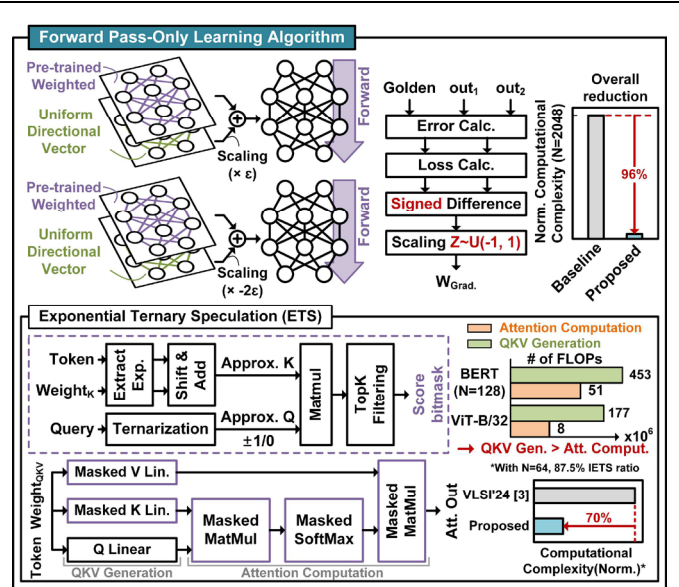


Fig. 2. Workflow of the forward pass only learning algorithm.

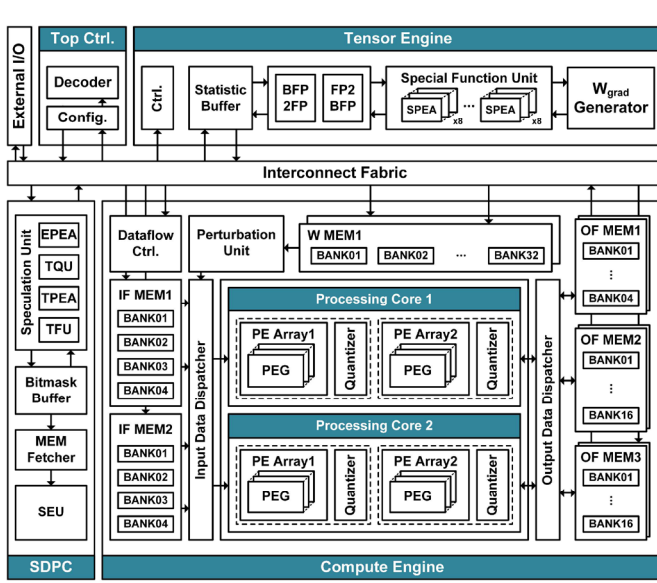


Fig. 3. Architecture of the proposed Transformer learning processor.

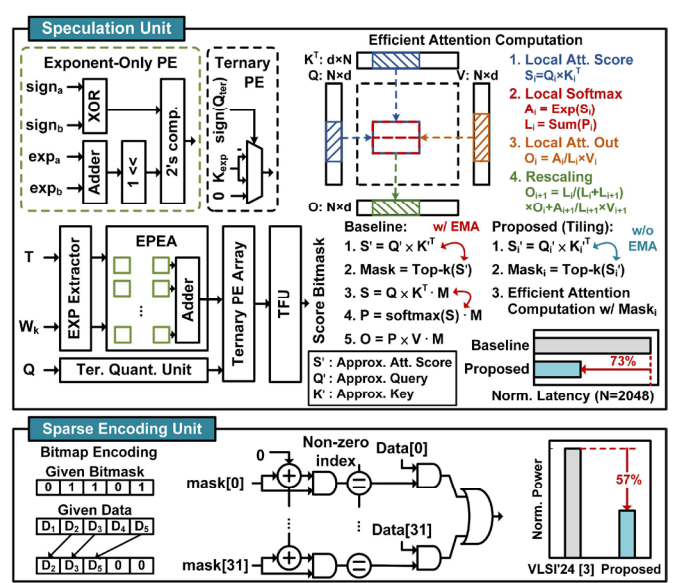


Fig. 4. Hardware details of the key sub-modules in the SDPC.

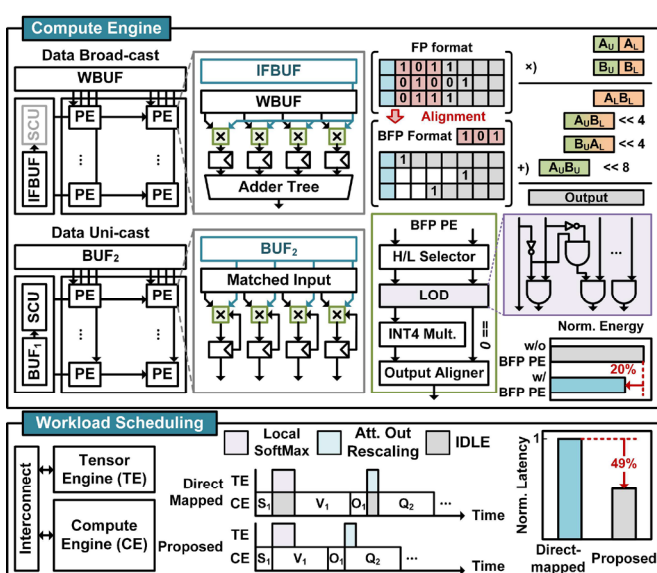


Fig. 5. Compute engine architecture and workload scheduling.

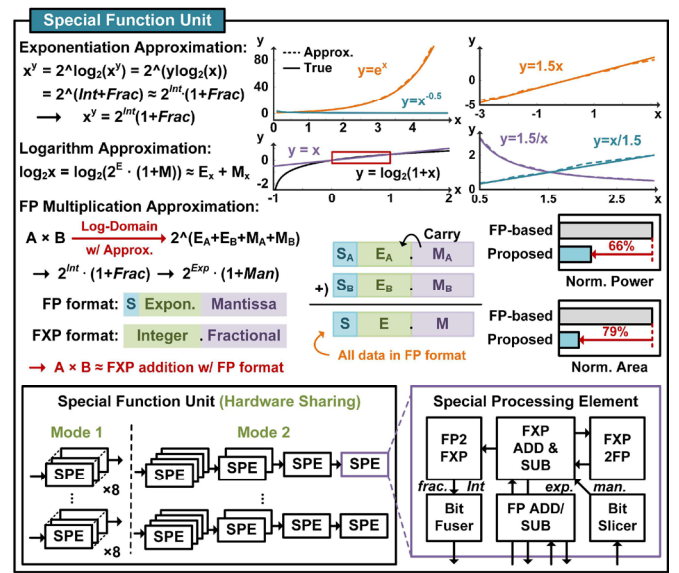


Fig. 6. Piecewise affine operations for the special function unit in the tensor engine.

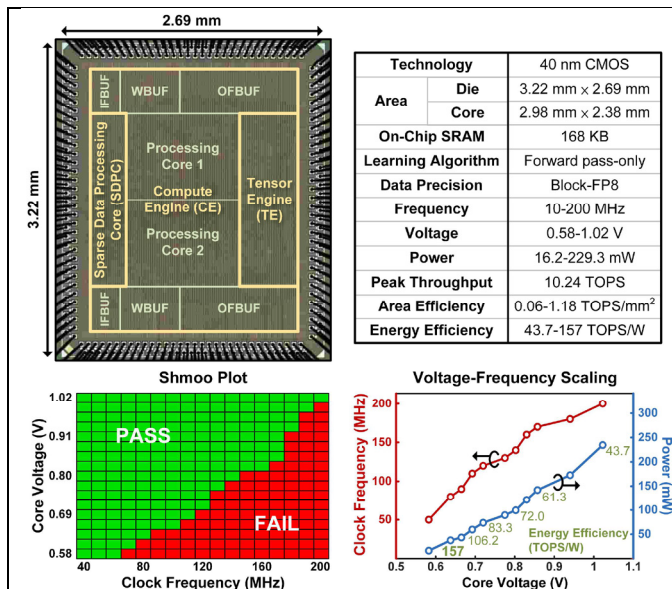
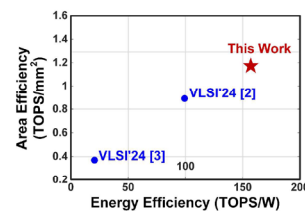


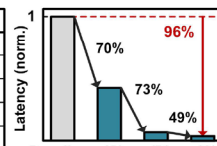
Fig. 7. Chip implementation and measurement results.

NLP Task	BP-Based	This Work	Performance Degradation
BERT			
SST-2	93.5%	92.6%	0.9%
MRPC	88.9%	88.0%	0.9%
CoLA	52.1%	52.0%	0.1%
RoBERTa			
SST-5	46.7%	45.7%	1.0%
SNLI	74.9%	73.7%	1.2%



	VLSI'24 [3]	VLSI'24 [4]	This Work
Technology	40 nm	22 nm	40 nm
Die Area (mm ²)	8.28	6.4	8.66
Learning Algorithm	BP-based	BP-based	FWP-only
Data Precision	Block-FP8	BF16, FP8	Block-FP8
Supply Voltage (V)	0.6-1.16	0.6-1.0	0.58-1.02 V
Frequency (MHz)	10-200	115-495	10-200
Power (mW)	9.7-119	49.2-461.2	16.2-229.3
Max Area Efficiency (TOPS/mm ²)	0.89	0.37 (BF16) 0.8 (FP8)	1.18*
Max Energy Efficiency (TOPS/W)	99.2	20.58 (BF16) 54.94 (FP8)	157*

*Measured with 87.5% IETS filtering ratio, a sequence length of 2048, and a tiling size of 64



- (A) Parallel processing
(B) Tile-based ETS[†] with efficient attention computation
(C) Workload scheduling

Fig. 8. Performance evaluation and comparison with prior arts.

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